

## ***Effect of the vegetarian diet on non-communicable diseases.***

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A vegetarian diet generally includes plenty of vegetables and fruits, which are rich in phytochemicals, antioxidants, fiber, magnesium, vitamins C and E, Fe<sup>3+</sup>, folic acid and n-6 polyunsaturated fatty acid (PUFA), and is low in cholesterol, total fat and saturated fatty acid, sodium, Fe<sup>2+</sup>, zinc, vitamin A, B<sub>12</sub> and D, and especially n-3 PUFA. Mortality from all-cause, ischemic heart disease, and circulatory and cerebrovascular diseases was significantly lower in vegetarians than in omnivorous populations. Compared with omnivores, the incidence of cancer and type 2 diabetes was also significantly lower in vegetarians. However, vegetarians have a number of increased risk factors for non-communicable diseases such as increased plasma homocysteine, mean platelet volume and platelet aggregability compared with omnivores, which are associated with low intake of vitamin B<sub>12</sub> and n-3 PUFA. Based on the present data, it would seem appropriate for vegetarians to carefully design their diet, specifically focusing on increasing their intake of vitamin B<sub>12</sub> and n-3 PUFA to further reduce already low mortality and morbidity from non-communicable diseases. © 2013 Society of Chemical Industry.